

Rithvij Pasupuleti

Lawrence, Kansas

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October 24th, 2025

Hiring Committee

The Robotics Department

University of Michigan

Dear Hiring Committee,

I am writing to express my enthusiasm for the Applications Programmer Associate position within the Robotics Department at the University of Michigan. With a Master's degree in Computer Science and over three years of professional experience developing and supporting enterprise-level web applications in the fintech, healthcare, and research domains, I am eager to contribute to Michigan Engineering's mission of advancing technology for the common good.

In my previous role at Cognizant, I developed and maintained scalable .NET and JavaScript based applications that supported critical business workflows. I collaborated closely with cross-functional teams to gather requirements, implement features, and troubleshoot production issues - experiences that strengthened my skills in coding, debugging, and database management using SQL Server and Entity Framework Core. I also integrated RESTful APIs, optimized backend performance and adhered to strict security and compliance guidelines through JWT and OAuth methodologies.

Beyond technical expertise, I take pride in being a proactive team player who values clear documentation, knowledge sharing, and continuous process improvement. The Robotics Department's commitment to innovation, collaboration, and technical excellence strongly aligns with my own passion for building impactful and reliable software that supports research and real-world problem-solving.

I would be honored to bring my technical background, problem-solving mindset and dedication to continuous learning to the Michigan Engineering community. Thank you for considering my application. I look forward to the opportunity to discuss how I can contribute to your team's success.

Sincerely,
Rithvij Pasupuleti

Rithvij Pasupuleti

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Summary

Results-driven software and application developer with over 3 years of experience building and deploying efficient software solutions across banking, healthcare and research domains. Adept at collaborating with cross-functional teams to deliver user-centric applications in dynamic, fast-paced environments. Passionate about continuous learning and innovation with expertise in microservices, object-oriented software architectures, RESTful API design and cloud technologies. Known for strong communication and problem-solving skills, and a commitment to driving efficiency in software systems.

EDUCATION

- **University of Kansas**, Lawrence, KS *Aug 2023 – May 2025*
Master's degree in Computer Science, **GPA: 4.0/4.0**
Coursework: Advanced Data Structures and Algorithms, Analysis of Algorithms, Software Security Evaluation, Machine Learning, Embedded ML, Operating systems, Deep reinforcement learning

SKILLS

- **Programming Languages:** JavaScript, TypeScript, C#, C++, Python, SQL, HTML, CSS
- **Frameworks & Technologies:** ASP.NET Core (MVC, Razor), AngularJS, React, Next.js, Node.js, Express.js, Pega (PCSA v8.6), Tailwind CSS, GraphQL, Ocelot API Gateway, Microservices, OOP, Event-Driven Architecture, JQuery, AJAX, RESTful APIs, XML, JSON
- **Cloud & DevOps:** AWS (Lambda, EC2, S3, SQS, SNS, CDK, CloudFormation), Azure (Azure App services, Blob Storage, Azure Service Bus, Event Grid), Docker, Git, GitLab, GitLab CLI, Swagger, Postman, CI/CD
- **Data & Infrastructure:** MySQL, MongoDB, Microsoft SQL Server, DynamoDB, Redis, Kafka, RabbitMQ, Linux Shell Scripting
- **AI/ML & Software Practices:** Langchain, RAG pipelines, LLM, Agentic AI, Generative AI, TensorFlow, PyTorch, Scikit-learn, Pandas, NLP, NLTK, SpaCy, BERT, Unit Testing, Integration Testing, Regression Testing, Grafana, Splunk, Sentry, Agile, Scrum, Kanban, SDLC

WORK EXPERIENCE

Cognizant Technology Solutions

Aug 2021 – July 2023

Programmer Analyst

- Engineered enterprise web applications for banking and healthcare, enhancing performance and user experience with JavaScript and C# frameworks.
- Designed modular Microservices in C# following Object Oriented SOLID principles, making core banking services easier to maintain and extend.
- Optimized front-end modules with AngularJS and React, resulting in faster load times (40% improvement) and more engaging user experiences (30% increase in engagement).
- Built and integrated RESTful APIs (JSON and XML) with ASP.NET Core, Node.js and Express.js, reducing data exchange latency and boosting API efficiency by 15%.
- Managed and structured database schema across MySQL and MongoDB, ensuring reliable handling of sensitive banking and healthcare data.
- Automated backend infrastructure using AWS CDK, including API Gateway and Lambda, streamlining third-party integrations for secure software workflows.
- Built and maintained CI/CD pipelines using GitLab and Docker, streamlining deployments and helping the team deliver features more efficiently
- Built low-code applications and automated business processes with Pega, streamlining and improving overall efficiency by 20%

- Developed efficient RAG pipelines using LangChain and ChromaDB, cutting document retrieval times by 40% and streamlining technical workflows.
- Applied AI and LLM tools (OpenAI, Llama) to automate repetitive data analysis and reporting, reducing manual effort by 50%.
- Built Python pipelines to efficiently process large-scale remote sensing imagery, improving data throughput by 30% for model training and analysis.
- Connected and Integrated multiple data sources through RESTful APIs (Microsoft, Sentinel Hub), creating a scalable pipeline for continuous and reliable data access
- Designed secure data environments using MongoDB, MS SQL Server and AWS S3, enforcing robust OAuth authentication and access controls
- Developed and fine-tuned deep learning models (SAM, VGG-16, U-Net) for complex image segmentation, improving overall prediction accuracy by 35%

PROJECTS

- **Full-Stack e-Commerce Platform**
Developed and deployed a [full-stack e-Commerce platform](#) using MERN stack (MongoDB, Express.js, React, Node.js), integrating Stripe and Razorpay for secure payment, enabling product browsing, cart and order management.
- **Distributed Microservices Application**
Built a modular [microservices platform](#) using ASP.NET Core, Ocelot API Gateway, Azure Service Bus, SQL Server, RabbitMQ, Docker, and ASP.NET Identity for authentication and authorization.
- **E-commerce Bookstore Website**
Developed a [full-stack e-Commerce bookstore](#) website using ASP.NET Core MVC, Razor Pages, C#, JQuery and Bootstrap. Integrated Stripe for secure payment, implemented order management, and deployed the website on Azure.
- **Modern Banking Application**
Designed a [modern banking application](#) using Next.js, React, TypeScript and Appwrite for backend, featuring seamless account linking via Plaid, real-time transactions, and an interactive finance dashboard with TailwindCSS and Chart.js.
- **LDPC Decoder for 5G NR**
Simulated and analyzed the performance of LDPC codes in 5G NR technology using Iterative Belief Propagation algorithm and Convolutional Neural Networks (BP-CNN), achieving improved bit error rate reduction in real-time message data.
Publication: Performance Analysis of optimal LDPC Decoder for 5G using Belief Propagation Algorithm at IEMECON 2021 (10th International Conference on Internet of Everything, Microwave Engineering, Communication and Networks)

Links : <https://github.com/Rithvij>